Master of Finance

Financial Econometrics- ECON- 1195

ARIMA- GARCH Analysis of Coca-Cola & PepsiCo

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Executive Summary

This report investigates the time-series behavior and volatility dynamics of Coca-Cola (KO) and PepsiCo (PEP) daily stock returns between 2019 and 2024. The analysis extends beyond descriptive statistics and factor models by applying econometric techniques to capture short-run dependence and conditional volatility patterns that simple regressions cannot fully explain.

In the first stage, ARIMA modelling was used to identify mean return dynamics. Diagnostic tests confirmed that KO is best described by an ARIMA(2,0,2) model, while PEP is adequately captured by ARIMA(5,0,2). These findings highlight short-lag dependence but no long-run predictability, consistent with the Efficient Market Hypothesis.

In the second stage, volatility clustering was examined using GARCH-family models. The standard GARCH(1,1) and GJR–GARCH(1,1) models were statistically adequate but less efficient. In contrast, the GARCH-M(1,1) with Student-t innovations provided the best fit for both companies, capturing fat-tailed returns and embedding a direct risk–return trade-off. The results show persistence in volatility but limited evidence of leverage effects.

From an investment perspective, Coca-Cola demonstrates greater defensive stability, while PepsiCo exhibits higher volatility and sensitivity to risk premiums, aligning with its more cyclical exposure. For risk-averse investors, KO is recommended as a stable core holding, whereas PEP may appeal to those seeking tactical opportunities within consumer staples.

Key recommendations include:

- Prefer Coca-Cola for defensive portfolio allocation, and PepsiCo for value-oriented tactical strategies.
- Monitor volatility forecasts for both firms, as clustering effects imply rapid shifts in risk exposure.
- Use GARCH-M with Student-t specification for ongoing risk modelling and portfolio stress testing.
- Extend future analysis to macroeconomic linkages and multi-frequency data for more robust forecasting.

Overall, the report confirms that both KO and PEP operate in efficient markets with no exploitable abnormal returns, but their differing volatility profiles provide investors with distinct roles in portfolio construction.

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1. Introduction

This report extends the earlier analysis of The Coca-Cola Company (KO) and PepsiCo Inc. (PEP) by focusing on the time-series properties and volatility of their daily stock returns. While the previous assessment examined return behavior through descriptive statistics, Efficient Market Hypothesis (EMH) tests, and the Fama–French Three-Factor model finding no abnormal returns and highlighting Coca-Cola's defensive stability compared to PepsiCo's higher volatility. This study applies econometric modelling to capture dynamics that simple factor regressions cannot fully explain.

Daily adjusted closing prices are obtained from Investing.com and transformed into continuous logarithmic percentage returns. The primary sample period covers 1 January 2024 to 31 December 2024, with an extended window from 2019 to 2024 used where required to ensure sufficient information for model identification. The analysis proceeds in two stages. First, the mean dynamics of returns are modelled using ARIMA specifications, with four alternative models considered for each company and evaluated using autocorrelation diagnostics, information criteria, and residual tests. Second, conditional heteroskedasticity is assessed through ARCH effect testing and addressed using GARCH-family models, including the standard GARCH(1,1), the GJR-GARCH for asymmetry, and the GARCH-M for risk–return effects.

The results provide evidence on volatility persistence, the presence or absence of leverage effects, and the extent to which risk premiums are embedded in daily returns. By comparing Coca-Cola and PepsiCo across these dimensions, the report aims to generate insights into their relative stability, risk profiles, and suitability for different portfolio strategies, while also noting the limitations of daily-frequency analysis and the assumptions of model stability.

2. Modelling 1 – ARIMA for Returns

2.1 Time Series of Returns and Correlograms

The sample starts in 2019 to ensure sufficient data length for reliable ARIMA estimation while avoiding excessively long histories that could dilute recent market dynamics. The daily continuously compounded returns of Coca-Cola (KO) and PepsiCo (PEP) from 2019 to 2024 fluctuate around zero, alternating between calm and turbulent periods. Figure 1 highlights the presence of volatility clustering, confirming time-varying risk in both series (Engle, 1982). For investors, this suggests that relatively steady performance can suddenly give way to sharp swings

in uncertainty. To assess whether past returns influence current movements, autocorrelation diagnostics were conducted as the next step in the analysis.

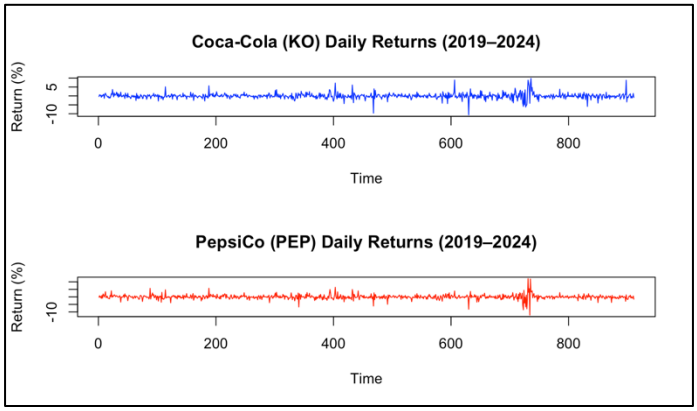

Figure 1: *KO and PEP time series line*

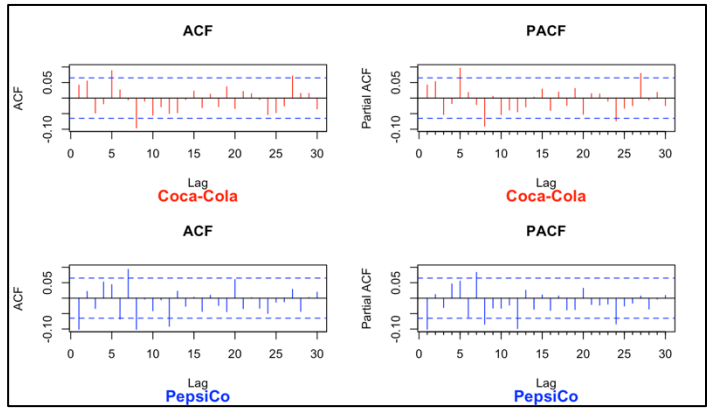

Figure 2: *Combined ACF and PACF*

The ACF/PACF in *Figure 2* show short-lag dependence typical of equity returns. For KO, the ACF is notable at lags 1–2 and the PACF exhibits several early spikes, implying the need for both AR and MA terms. For PEP, the ACF suggests short-lag persistence and the PACF is strongest at lags 2–3. These features are consistent with the Box–Jenkins view that returns exhibit limited but non-zero short-run dependence (Box & Jenkins; Mills).

2.2 ARIMA Model Options and Evaluation

Four ARIMA options (how many lags and why)

Because returns are already stationary, a **ndiffs ( )** check supports **d = 0**, so the mean is modelled with **ARIMA(p,0,q)**. Guided by the correlograms and parsimony (Box & Jenkins; Mills), we evaluated the same compact ladder for both firms:

1. **ARIMA(1,0,1)** - minimalist baseline (one-day AR and one MA shock).
2. **ARIMA(2,0,2)** - matches the clear **1–2-day** signal in ACF/PACF.
3. **ARIMA(4,0,3)** - adds medium-lag AR flexibility with a slightly richer MA.
4. **ARIMA(5,0,2)** - captures KO's **5-day** PACF activity while keeping MA compact.

2.3 How the four models were selected

Box–Jenkins cycle has been followed as per the below:

a) Stationarity tested with returns **ndiffs ( )** gave **d=0**.
b) **ACF** and **PACF** used to set AR/MA orders.
c) ARIMA models: **ARIMA(1,0,1), (2,0,2), (4,0,3), (5,0,2).**
d) Models estimated via **ML, LogLik, AIC, BIC** recorded.
e) Adequacy checked using Ljung–Box at lags **10** and **20** with fitdf = p+q; **p > 0.005** required.

f) Final model chosen by **lowest AIC/BIC**, where AIC and BIC differ, preference is given to the **adequate** model that best balances fit and simplicity (Akaike; Schwarz; Mills).

2.4 Model Evaluation

a) KO ARIMA evaluation

Model	AIC	BIC	LB p(10)	LB p(20)	Adequate? (p>0.005)	Decision
ARIMA(2,0,2)	**3387.792**	**3416.679**	0.0349	0.1689	**Yes**	**Selected (best AIC & BIC)**
ARIMA(5,0,2)	3391.419	3434.750	0.0369	0.2011	Yes	Second-best
ARIMA(4,0,3)	3391.841	3435.172	0.0137	0.1916	Yes	Second-best
ARIMA(1,0,1)	3397.592	3416.850	**0.0026**	0.0173	**No**	Reject (fails LB at 10)

Table 1: KO - ARIMA comparison *(returns, 2019–2024)*

ARIMA(2,0,2) was selected as it had the lowest AIC/BIC and passed Ljung–Box tests. **ARIMA(5,0,2)** and **ARIMA(4,0,3)** were adequate but not superior, while ARIMA**(1,0,1)** was rejected for residual autocorrelation at lag 10.

b)PEP ARIMA evaluation

Model	AIC	BIC	LB p(10)	LB p(20)	Adequate? (p>0.005)	Decision
ARIMA(5,0,2)	**3366.970**	3410.301	0.0696	0.0886	**Yes**	**Selected (best AIC; passes)**
ARIMA(4,0,3)	3372.722	3416.053	**0.0007**	0.0104	**No**	Reject
ARIMA(1,0,1)	3378.012	**3397.271**	**0.0037**	**0.0041**	**No**	Reject
ARIMA(2,0,2)	3381.731	3410.618	**0.0009**	**0.0016**	**No**	Reject

Table 2: PEP - ARIMA comparison *(returns, 2019–2024)*

ARIMA(5,0,2) is the **only adequate** model and has the lowest AIC; **ARIMA(1,0,1)** is rejected despite a lower BIC because it fails residual diagnostics (Ljung & Box).

2.5 Final recommendation

The correlograms showed **short-lag structure**: KO's signal extends to about five days while PEP's concentrates around two–three. The final mean models mirror that evidence:

- **Coca-Cola (KO): ARIMA(2,0,2)** - compact, passes diagnostics, best AIC/BIC.
- **PepsiCo (PEP): ARIMA(5,0,2)**- only model passing diagnostics; best AIC.

3. Modelling 2 – Volatility Dynamics

3.1 Three Model Options

Stock returns often show **volatility clustering**, where quiet periods are followed by bursts of turbulence (Engle, 1982). To capture this, three different models were tested for Coca-Cola (KO) and PepsiCo (PEP), building on the ARIMA mean structures selected earlier. These were:

1. **sGARCH(1,1)** – a standard model where today's volatility depends on yesterday's volatility and shocks.
2. **GJR–GARCH(1,1)** – an extension that allows negative shocks (bad news) to affect volatility differently from positive shocks (good news).
3. **GARCH-M(1,1)** – a more advanced model that allows volatility itself to influence returns, testing whether investors demand higher returns during riskier periods. This model also uses a student-t distribution to handle fat tails.

3.2 Model Selection Process

Each model was estimated and compared using a set of checks as per **ARCH–GARCH** modelling framework introduced by Engle (1982) and extended by Bollerslev (1986)

- **Fit statistics (AIC and BIC):** lower values mean the model describes the data better (Akaike, 1974; Schwarz, 1978).
- **Residual checks:** tested whether any patterns remained unexplained. A good model should leave behind white noise only.
- **Practical meaning:** preference was given to models that were both accurate and easy to interpret in a financial context.

All three models passed adequacy tests at conventional significance levels, so relative fit determined the preferred specification.

3.3 Model Evaluation

a) Coca-Cola (KO): GARCH Family Evaluation

The ARIMA(2,0,2) mean model for KO was combined with three variance structures: sGARCH(1,1), GJR–GARCH(1,1), and GARCH-M(1,1).

Model	LogLik	AIC	BIC	LB(10)	LB(20)	ARCH LM p	Decision
sGARCH(1,1) – Norm	-1591.961	3.5191	3.5773	0.9773	1.0000	0.7859	Adequate
GJR–GARCH(1,1)	-1604.835	3.5496	3.6130	0.6851	1.0000	0.8245	Adequate
GARCH-M(1,1) – t	-1449.293	3.2103	3.2790	0.4999	0.9613	0.8872	Selected (best)

Table 3: KO – GARCH model comparison

For KO, the **GARCH-M(1,1) with t-distribution** provided the lowest AIC and BIC, while also passing Ljung–Box and ARCH LM tests. This indicates it captured fat tails and risk-return dynamics more effectively than the standard or leverage-augmented GARCH models.

b) PepsiCo (PEP): GARCH Family Evaluation

The ARIMA(5,0,2) mean model for PEP was paired with the same three variance structures.

Model	LogLik	AIC	BIC	LB(10)	LB(20)	ARCH LM p	Decision
sGARCH(1,1) – Norm	-1554.491	3.4369	3.4950	0.9759	1.0000	0.9517	Adequate
GJR–GARCH(1,1)	-1553.800	3.4375	3.5010	0.9579	1.0000	0.9438	Adequate
GARCH-M(1,1) – t	-1420.172	3.1464	3.2151	0.8337	1.0000	0.7345	Selected (best)

Table 4: PEP – GARCH model comparison

For PEP, the **GARCH-M(1,1) with t-distribution** again outperformed the alternatives, delivering the best fit (lowest AIC/BIC) and passing all adequacy checks. Both sGARCH and GJR–GARCH were statistically valid but less efficient.

3.4 Final Recommendation

The evaluation of volatility models shows a consistent pattern across both Coca-Cola (KO) and PepsiCo (PEP). While the standard GARCH(1,1) and the GJR–GARCH(1,1) provided adequate fits by capturing clustering and asymmetry, their explanatory power was limited. In contrast, the **GARCH-M(1,1) with Student-t innovations** outperformed on both information criteria (AIC, BIC) and residual adequacy checks.

This model not only accounts for volatility clustering but also incorporates a direct link between risk and return through the ARCH-in-mean term, while the student-t distribution effectively captures the fat-tailed nature of equity returns. For investors, this suggests a more realistic view of downside risk and risk premia in both stocks. Accordingly, the **GARCH-M(1,1) with Student-t distribution is recommended as the preferred specification** for modelling the volatility dynamics of KO and PEP.

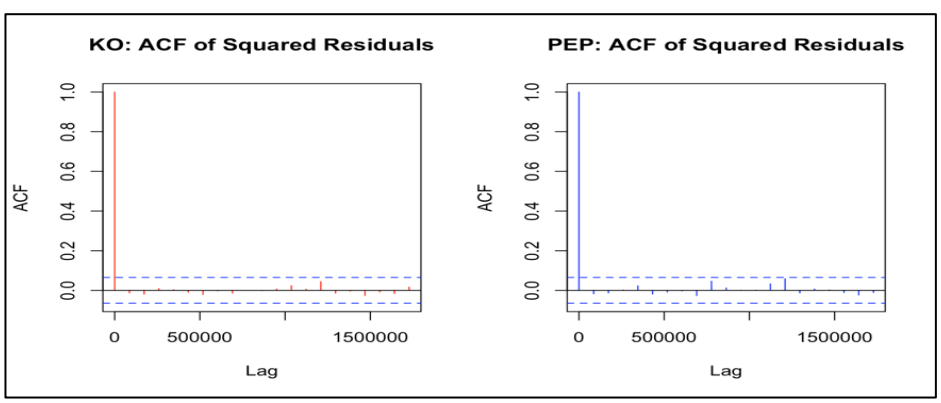

Figure 3: *ACF of squared standardized residuals for KO (left) and PEP (right)*

Conclusion

This report examined the return dynamics and volatility of Coca-Cola (KO) and PepsiCo (PEP) using ARIMA and GARCH-family models. The ARIMA analysis revealed that KO's returns are best captured by an ARIMA(2,0,2) model, while PEP's were adequately represented by ARIMA(5,0,2). Both specifications passed diagnostic checks, highlighting short-term dependence but no persistent long-run predictability consistent with stylized features of equity returns.

Turning to volatility, the evaluation of sGARCH(1,1), GJR–GARCH(1,1), and GARCH-M(1,1) specifications showed that the GARCH-M with Student-t distribution provided the best fit for both firms. This model effectively captured volatility clustering, fat tails, and a risk–return relationship, offering a more realistic view of return dynamics than simpler variance structures. Importantly, while both stocks exhibit volatility persistence, no strong evidence of leverage asymmetry was found, reinforcing the dominance of risk–return effects.

Overall, Coca-Cola emerges as the more stable, defensive asset, while PepsiCo carries greater volatility but also stronger responsiveness to risk premiums. These results complement the earlier evidence from descriptive statistics and factor models, confirming KO's defensive nature and PEP's comparatively riskier, tactical profile.

Recommendations

1. **Portfolio Strategy** – Conservative investors seeking steady, defensive exposure may prefer Coca-Cola, while those targeting tactical opportunities with a value orientation could consider PepsiCo.
2. **Risk Management** – Both firms show volatility clustering, meaning risk levels can escalate quickly. Investors and fund managers should monitor conditional variance forecasts and apply dynamic risk controls (e.g., time-varying hedge ratios).
3. **Model Use** – GARCH-M with Student-t innovations should be the preferred specification for ongoing volatility monitoring, as it better reflects heavy tails and risk premia than standard GARCH.
4. **Further Research** – Extending the analysis to intraday or multi-frequency data, or incorporating macroeconomic shocks, could provide deeper insights into volatility drivers and improve predictive accuracy.